

7V (poor light) to 20-21V under open circuit with bright light. The panel gives its highest power at a voltage, V_p , that varies between, say, 13.5V and 18V depending on the cell efficiency. Most of the loads, however, need constant voltage.

A PV panel is basically a current source and the current level depends on the intensity of the incident sunlight. That's why for harvesting maximum energy from a PV panel, it is essential to use a current controller to ensure that the panel delivers its peak potential energy under all light conditions.

In case one is using a solar panel to charge a battery, the use of a current controller is a must. Without it, the user will damage the connected battery. The most widely used current controller uses a microcomputer that tracks peak energy delivery point of the panel or an array all the time and ensures that the associated DC-DC converter delivers maximum peak power (MPPT).

How does it help to improve the solar power output?

As described above, MPPT type controller is a must for any serious application of solar power above, say, 50Wp. The controller cost makes it unadvisable to use it below that power. For lower-power panels, a simple switch-type controller is used. It ensures that the connected battery or the other load does not get damaged.

What's an embedded solar panel?

In an embedded solar panel (ESP), under a process that is awaiting the grant of a patent, the solar energy generated by a solar panel of any type is processed such that it is harvested to its full potential under any solar condition. This energy processor delivers the energy over a range of high voltage up to 300V so that all panels have identical performance.

In what way does it improve the efficiency and lower the cost?

There are several ways in which ESPs improve the power plant's efficiency. At the panel level, the embedded circuit of an ESP ensures harvesting of peak power from the host panel using a unique MPPT algorithm and delivers it any voltage that the load requires. The best advantage of ESPs is that they allow an array designer to use 'all parallel' architecture because of its capacity to connect hundreds of panels in parallel, each panel delivering its peak power into a bus bar. Since each panel delivers power at high voltage, the array designer can avoid completely the series strings and expensive work to select and match panels. Such a parallel array of ESPs helps to harvest 10 to 12 per cent more energy than the conventional configuration currently used. Ten per cent more of free energy will change the ROI of every solar plant in the world.

What all are the ideal applications for it?

Any power plant from a 200Wp to a several megawatt capacity can use ESPs cost-effectively.

Currently, what are the major solar systems/devices in use in India?

As described above, all current installations, except a few in the US, use PV panel arrays in a conventional series-parallel architecture.

Which segments do you see picking up and why?

Till the subsidies, as in the western countries, are put in place, small applications that provide backup power during blackout, terrace-mountable systems for urban use and telecom applications for field equipment powering will not grow as rapidly as these should.

The absence of government subsidy for solar plants below a megawatt doesn't make sense. As of today, government policy is to buy solar power from bulk producers having megawatt-size installation. The government plans to pay these large investors Rs 18 per unit. In comparison, a 10kW rooftop power plant by a housing society or a group of rural homes will get nothing for the units of power it would generate. This makes no sense if the government is serious about using solar power as per its ambitious plans.

In Germany and most other countries, even a small producer of solar power can feed it into the grid. A bidirectional metering keeps account of what is fed back into the grid, and for what is fed the power plant owner gets higher price per unit (Rs 18 in India).

Further, even in case of large plants, it is not just the question of current government buying the grid power at Rs 18 per unit but also a guarantee from the government that this rate will be valid for entire 20-year life of the plant as the ROI will entirely depend on the government subsidy being available throughout the plant life.

Such a price per unit will especially help rural households. If a farmer family produces its own need for one unit of power per day, it will need to install a 200W panel. If that farmer is paid Rs 18 per day for the power he generates on his own, he can easily pay EMI for his solar power plant even if he gets his loan at 12 per cent from a bank.

The government will actually benefit if such one-unit-a-day solar panels are distributed free instead of giving subsidy per unit.

What can we expect in solar power technology in the near future?

There is little chance to find an alternative to photovoltaic solar panels for a direct conversion of light into electrical power. Over the next five to seven years, PV solar will prevail. Current power conversion efficiency of 17 per cent will go up to about 24 per cent with process innovation. Solar thermal power plants look like a good supplement to PV solar.